



Batch / Lot ID: BATCH_20260914_233438	Inspection Date: 2026-09-14 23:34:38
Procurement Center: CENTER_DL_001	Quality Inspector: OPERATOR_DEMO_01 (Certified)
GPS Geotag: 28.7041° N, 77.1025° E (Delhi)	AI Engine: YOLOv8-Seg (Dense) + MobileNetV3 (98.5% Acc)

97.67% GRADE A RATIO 42 of 43 units	2.33% URS DEFECT RATIO 1 rejected units	43 TOTAL DETECTED 100% Instance Mapped	PASSED LOT (ACCEPTED) Mandatory Limit ≥ 75%
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1. Annotated Batch Image Snapshot & Instance Bounding Boxes

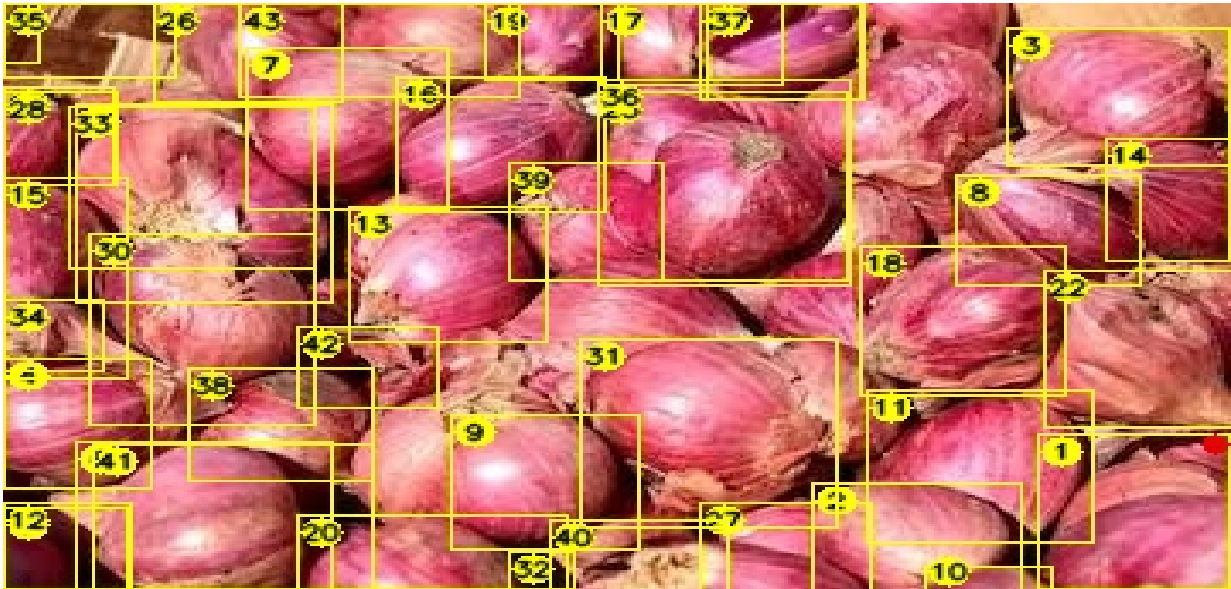


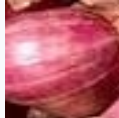
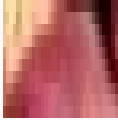
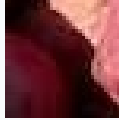
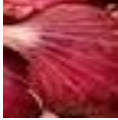
Figure 1: High-contrast yellow square bounding boxes and numbered index badges identifying all 43 detected onion bulbs.

2. Defect Category Breakdown & Procurement Tolerance Analysis

Defect Category	Count (Units)	Percentage (%)	NAFED Mandi Tolerance	Compliance
Grade A	42	97.7%	Base Grade	Target ≥ 75%
Damaged	1	2.3%	Max 5.0%	Within Tolerance
Rotten	0	0.0%	Max 2.0%	Within Tolerance
Sprouted	0	0.0%	Max 3.0%	Within Tolerance
Undersized	0	0.0%	Max 15.0%	Within Tolerance

3. Individual Onion Inspection Thumbnails & Defect Review Gallery

Isolated sub-image crops extracted directly from the batch scan for optical verification, AI classification, and supervisor audit.

 #1 Damaged Conf: 99% Size: 2.52cm PASSED	 #2 Grade A Conf: 100% Size: 2.44cm PASSED	 #3 Grade A Conf: 100% Size: 2.6cm PASSED	 #4 Grade A Conf: 100% Size: 2.12cm PASSED
 #5 Grade A Conf: 100% Size: 1.44cm PASSED	 #6 Grade A Conf: 100% Size: 3.0cm PASSED	 #7 Grade A Conf: 100% Size: 2.64cm PASSED	 #8 Grade A Conf: 99% Size: 2.16cm PASSED
 #9 Grade A Conf: 100% Size: 2.2cm PASSED	 #10 Grade A Conf: 100% Size: 1.48cm PASSED	 #11 Grade A Conf: 100% Size: 2.64cm PASSED	 #12 Grade A Conf: 99% Size: 1.48cm PASSED
 #13 Grade A Conf: 100% Size: 2.32cm PASSED	 #14 Grade A Conf: 99% Size: 2.0cm PASSED	 #15 Grade A Conf: 100% Size: 3.28cm PASSED	 #16 Grade A Conf: 100% Size: 2.44cm PASSED

4. Certified Operator Review & Override Trail

Timestamp	Onion #	Original Classification	Overridden Class	Reviewer ID
2026-09-14 23:34:38	#1	Grade A	Damaged	SUPERVISOR_01

CRYPTOGRAPHIC AUDIT SEAL
SHA256: BATCH_20260914_233438-DOCA-NAFED-43-PASS
Tamper-evident verification key logged to Central Mandi Repository.

OFFICER SIGN-OFF
OPERATOR_DEMO_01
Certified Quality Inspector, Ministry of Consumer Affairs